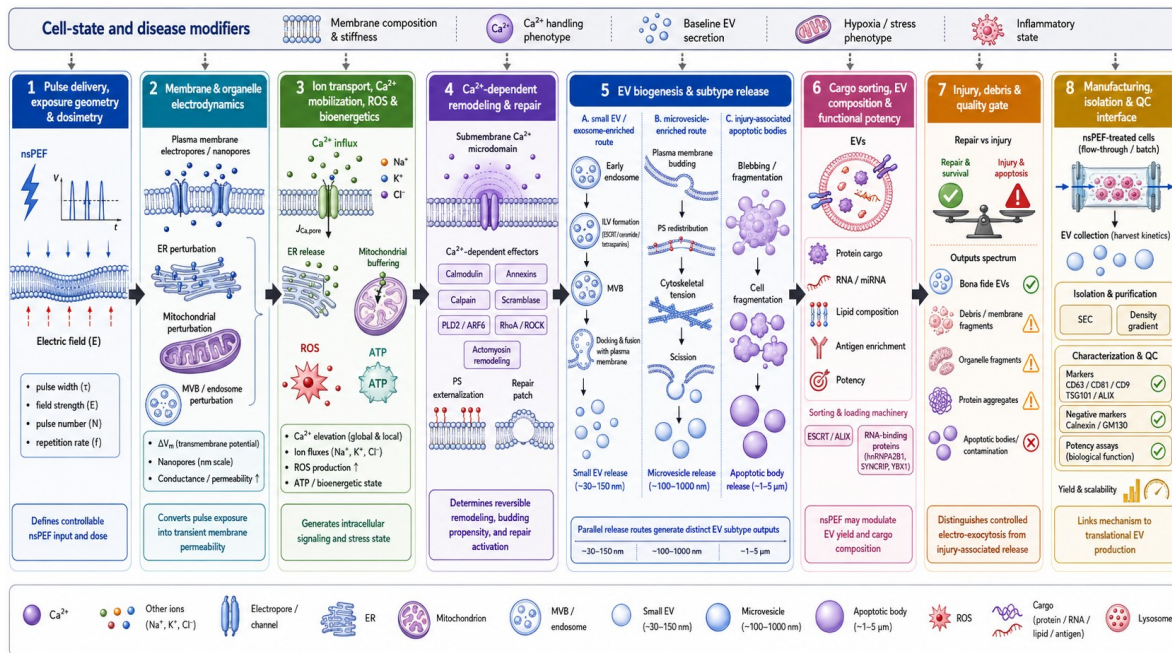


COURSE PLANNING GUIDE

From Cell Physiology to Modular EV Biogenesis Simulation

A beginner-first computational systems biology course built around electro-exocytosis

Source review: BIEN5700 Systems Physiology, BIEN6931 Modeling and Simulations of Integrated Cellular Systems, the EV-biogenesis manuscript, and the executable electroexo model



The course starts with familiar cell physiology and modeling patterns, then earns its way to this integrated nsPEF-to-EV framework.

Prepared for long-range course development | August 2026

1. Executive recommendation

Recommended structure: Use a 14-week beginner-first course for the full offering, and retain the existing 10-session slide sequence as a compressed seminar version. The full EV framework should appear only after students can explain a compartment, a mass balance, a membrane voltage, and an exposure-response chain in plain language.

The reviewed materials support a coherent progression from physiology to modeling and then to the EV-biogenesis application. The strongest teaching spine is not the order of the manuscript. It is a repeated sequence: biological question, system boundary, stocks and flows, governing balance, simulation, and interpretation.

Course block	Weeks	Purpose
I. Biological and modeling grammar	1-7	Homeostasis, transport, compartments, ODEs, membrane voltage, calcium, and rate laws.
II. Transferable modeling templates	8-9	PK/PD, PBPK/PD, physiological flows, variability, and model credibility.
III. EV-biogenesis capstone	10-14	Walk through the eight electroexo layers, integrate scenarios, and evaluate evidence and limits.

Course blocks are synthesized from the two course syllabi, the manuscript module map, and the current repository examples.

- Start with homeostasis and transport, not with code or the manuscript equations.
- Introduce ODEs through mass balance and visual stock-flow diagrams; analytical solution methods are optional enrichment.
- Use PK/PD and PBPK/PD as modeling templates for staged inputs, compartments, delayed effects, and physiological flows.
- Teach resting membrane potential and externally induced transmembrane voltage as related but distinct concepts.
- End every module with an input-state-output-evidence summary so students see how modules connect.

Terminology note: This plan interprets 'PBPD' as PBPK/PD. That matches the available PBPK_spring_2018.pdf and the standard progression from pharmacokinetics/pharmacodynamics to physiology-based models. If a different PBPD formulation was intended, it can be substituted in Week 9 without changing the course architecture.

2. Audience, outcomes, and teaching method

Audience and prerequisites

The target audience is senior undergraduates, graduate trainees, clinicians, or life scientists who understand basic cell biology but have little or no computational background. Algebra, graph reading, and willingness to edit a few parameters are sufficient at entry; calculus and programming are introduced as tools rather than gatekeeping prerequisites.

Learning outcomes

- Define a biological system boundary and distinguish inputs, state variables, parameters, and outputs.
- Translate a stock-flow diagram into a mass-balance equation and explain every term biologically.
- Explain when a well-mixed compartment is useful and when spatial gradients or microdomains matter.
- Simulate simple ODE models and interpret time courses, steady states, thresholds, and sensitivity.
- Explain Nernst, GHK, membrane capacitance, channel gating, and induced membrane charging without requiring a full Hodgkin-Huxley derivation.
- Use PK/PD and PBPK/PD as analogies for staged exposure, distribution, effect, and physiological flow.
- Trace an nsPEF protocol through the eight EV-biogenesis model layers to yield, subtype, cargo, potency, purity, and viability.
- Critique calibration evidence, identify placeholder assumptions, and propose a focused validation or model-improvement project.

A repeated six-step lesson pattern

1. Begin with one biological question that can be answered in words.
2. Draw the boundary and identify the compartments or pools.
3. Label inflows, outflows, production, consumption, and feedback.
4. Write the minimum governing equation and define units.
5. Run or inspect one simulation, then perturb one parameter.
6. Interpret the figure, name the assumption, and identify what evidence would test it.

3. Source review and chapter map

The source folders contain substantial duplication from backups. The tables below identify the canonical lecture blocks that are most useful for this course and the page ranges that should be adapted. Page numbers refer to PDF pages in the named file, not textbook pages.

BIEN5700 Systems Physiology: high-value background

Source	Pages	Recommended teaching use
Syllabus sp18.pdf	p. 2	Use the lesson-to-Guyton chapter map: Homeostasis ch. 1; membrane transport ch. 4; membrane potential ch. 5; muscle and neurotransmission chs. 6-8; hemodynamics ch. 14; exchange chs. 15-16; kidney chs. 25-29; metabolism chs. 68-70; energetics chs. 73-74; hormones ch. 75.
01homeostasis.pdf	pp. 6-12	Core: homeostasis, dynamic steady state, negative and positive feedback.
01homeostasis.pdf	pp. 13-41	Core: membrane structure, diffusion, electrical and pressure gradients, facilitated diffusion, osmosis, active transport, endocytosis, and exocytosis.
01homeostasis.pdf	pp. 42-59	Core: Nernst and GHK intuition, Na/K pump, action potential, voltage clamp, and patch clamp. Pages 60-65 are optional conduction examples.
02muscle.pdf	pp. 33-48	Support: neurotransmitter vesicles, depolarization, Ca ²⁺ release, excitation-contraction coupling, and ATP-dependent recovery.
03circ.pdf	pp. 18-38	Support for PBPK/PD: pressure-flow relations, vascular compartments, capillary exchange, and transport.
04kidney.pdf	pp. 2-4; 40-41	Support for compartment volumes and clearance. The detailed nephron sequence is optional.
06respir.pdf	pp. 30-45	Optional case study for diffusion, exchange surfaces, transport capacity, and coupled compartments.
10metab.pdf	pp. 1-10; 17-24	Support for glycolysis, citric acid cycle, electron transport, ATP, energetics, and physiological stress.
11endo.pdf	pp. 1-7; 30; 44	Support for receptor signaling, second-messenger amplification, and feedback control.

Reviewed root: /Users/aghorban/Library/CloudStorage/OneDrive-Personal/Documents/04-Universities/MCW_MU/Courses/BIEN5700-SystemPhysiology

BIEN6931 Computational Systems Biology: core modeling sequence

Source	Pages	Recommended teaching use
BIEN_6391_Spring_2018_Syllabus1.pdf	p. 2	Use modules 1-5 and 7 as the conceptual backbone. Cell-cycle modeling is a possible elective, not a prerequisite for EV biogenesis.
Dash_Lect1_Intro_Sim_Biol_Syst_Spring_2018.pdf	pp. 2-6; 27-37	Core: systems/computational biology, model purpose, iterative modeling, formulation, parameterization, validation, sensitivity, and communication.

Source	Pages	Recommended teaching use
Introduction to Simulation of Biological Systems.pdf	pp. 1-20	Core case study: model-building workflow and mechanistic versus nonmechanistic modeling through the aquarium example.
Introduction to Compartmental Modeling (1).pdf	pp. 1-11	Core: well-mixed assumption, amount-concentration-volume relations, one-compartment washout, reactions with flow, and two-compartment exchange.
Compartmental_modeling_supplement_spring_2018.pdf	pp. 1-15	Support: transport mechanisms, membrane fluxes, drug absorption/elimination, and hemodialysis.
Differential equations_Spring_2018.pdf	pp. 1-6	Core ODE vocabulary and initial conditions. Laplace transform pages 7-12 are optional enrichment.
Numerical_methods_Spring_2018.pdf	pp. 1-11; 15-18	Core: Euler, Runge-Kutta intuition, errors, convergence, and MATLAB ODE solvers. Stability pages 12-14 are instructor-level or advanced.
Introduction to Law of Mass Action.pdf	pp. 1-5	Core rate-law vocabulary and units for reversible and irreversible reactions.

Reviewed canonical archive root: /Users/aghorban/Library/CloudStorage/OneDrive-Personal/Documents/04-Universities/MCW_MU/Courses/BIEN6931-TopicsInBiomedicalEngineering/end of the semester back up/BIEN 6931 701 Topics in Biomedical Engr - 5132018 - 1103 PM

BIEN6931: application templates and advanced bridge

Source	Pages	Recommended teaching use
Compartmental_Kinetic_PKPD_Modeling.pdf	pp. 2-12; 24-29; 33-55	Core: dose-to-effect stages, direct/inverse problems, one- and two-compartment models, absorption, half-life, volume of distribution, and nonlinear kinetics. Pages 13-15 add population variability.
PBPK_spring_2018.pdf	pp. 1-15	Core: physiology-based compartments, blood flow, permeability-surface area exchange, organ balances, and applications.
Facilitated_Diffusion.pdf	pp. 1-6	Core bridge from membrane biology to saturable carrier models.
Electrochemical_gradient_4_2_2018.pdf	pp. 1-6	Core: Fick, Nernst-Planck, GHK flux, and charged-solute compartment balances.
Resting_membrane_potential_4_2_2018.pdf	pp. 1-10	Core: ion gradients, Nernst, resting potential, pump contribution, and voltage-gated channels.
Dash_Modeling Cellular Electrophysiology(1).pdf	pp. 2-19	Core/bridge: membranes, Nernst, current-voltage relations, gating variables, HH structure, GHK, and cardiac Ca ²⁺ context. Pages 20-22 are advanced Markov-channel modeling.
Dash_The Hodgkin-Huxley Model.pdf	pp. 1-10	Optional advanced reading. Use for structure and gating intuition, not as a prerequisite derivation.
Dash_Modeling Biochemical Reaction Systems.pdf	pp. 4-20	Core: mass-conservation matrix, mass action, Gibbs energy, energy coupling, ion transport, enzymes, and Michaelis-Menten. Pages 21-31 are advanced reversible/cooperative/inhibition models.
Regression_analysis_handout_LM_Spring_2018.pdf	pp. 1-8	Guided advanced lab: nonlinear regression, local minima, Levenberg-Marquardt, parameter uncertainty, and example fits.

Source	Pages	Recommended teaching use
Model_assessment.pdf	p. 1	Core checklist for fit and parameter assessment.

The scanned handwritten thermodynamics, enzyme-kinetics, and voltage-gating notes are better retained as instructor references; the clean Dash lecture decks cover the same concepts more accessibly.

Archive curation decisions

Decision	Materials
Use as the student spine	01homeostasis; Intro Simulation; Intro Compartmental Modeling; ODEs/numerics; PKPD; PBPK; electrochemical gradient/resting potential; biochemical reactions; the electroexo examples.
Adapt selectively	Muscle/neuro Ca ²⁺ and exocytosis; circulation/kidney/respiratory case studies; full cellular electrophysiology; regression and model assessment.
Defer or make optional	Laplace-transform derivations, exact eigenvalue solutions, full HH derivation, Markov-channel detail, cell-cycle module, whole-organ survey topics unrelated to the EV narrative.
Instructor-only references	Handwritten scanned thermodynamics, simple-enzyme, and voltage-gating notes; backup ZIP duplicates; exam solutions and student submissions.

This curation prevents the beginner course from inheriting the mathematical prerequisite load of the original graduate course.

4. Recommended 14-week course sequence

Each week assumes one concept-focused session plus one guided activity or lab. For a lecture-only format, the lab column becomes an in-class figure interpretation or parameter-prediction exercise.

Wk	Theme	Core concepts	Activity/lab	Source anchors
1	Why model living systems?	Homeostasis, feedback, scales, system boundaries, models as useful simplifications.	Draw a pulse-to-cell concept map before seeing the formal EV framework.	BIEN5700 01homeostasis pp. 6-12; Dash Intro pp. 2-6, 27-37.
2	Cells as compartments	Membranes, organelles, transport, exocytosis, well-mixed assumptions, stocks and flows.	Turn a generic cell into cytosol, ER, mitochondria, membrane, and extracellular compartments.	01homeostasis pp. 13-41; Intro Compartmental pp. 1-3.
3	Mass balance and ODEs	Amount vs concentration, inflow/outflow, production/consumption, state variables, initial conditions.	Build and interpret a one-compartment washout and a two-compartment exchange model.	Intro Compartmental pp. 1-11; Differential Equations pp. 1-6.
4	Numerical simulation	Euler and Runge-Kutta intuition, solver output, error, time step, plots, reproducibility.	Compare Euler with solve_ivp; explain when two curves disagree.	Numerical Methods pp. 1-11, 15-18; Intro Simulation pp. 1-20.
5	Membrane potential simulation	Ion gradients, Nernst, GHK, pumps, capacitance, current versus voltage.	Calculate single-ion equilibrium potentials, then vary permeabilities in a simple GHK model.	01homeostasis pp. 42-59; Electrochemical Gradient pp. 1-6; Resting Potential pp. 1-10.

Wk	Theme	Core concepts	Activity/lab	Source anchors
6	Channels, calcium, and exocytosis	Voltage gating, Ca ²⁺ as signal and stressor, ER stores, synaptic release, patch-clamp interpretation.	Annotate an action-potential/Ca ²⁺ figure and identify which terms belong in a reduced ODE model.	O2muscle pp. 33-48; Cellular Electrophysiology pp. 2-19; HH optional.
7	Biochemical rate laws	Mass action, reversible reactions, saturation, Michaelis-Menten, Hill activation, energy coupling.	Build a two-state reaction and a Hill switch; connect them to repair or secretion.	Law of Mass Action pp. 1-5; Biochemical Reaction Systems pp. 4-20; Endocrine pp. 1-7.

Weeks 1-7 establish the biological and mathematical grammar used repeatedly in the EV model.

Wk	Theme	Core concepts	Activity/lab	Source anchors
8	PK/PD as a template	Dose-exposure-effect, one/two compartments, absorption, half-life, volume, nonlinear elimination, E _{max} /Hill effect.	Simulate a one-compartment PK model and map concentration to effect.	PKPD pp. 2-12, 24-29, 33-55.
9	PBPK/PD and credibility	Physiological compartments, blood flow, permeability, clearance, variability, fitting, sensitivity, and limits.	Extend to two organs or compartments; perturb flow/permeability; write a model-credibility note.	PBPK pp. 1-15; circulation pp. 18-38; kidney pp. 2-4, 40-41; Model Assessment.
10	Layer 1: pulse and dosimetry	Pulse amplitude, width, number, repetition rate, waveform, geometry, conductivity, energy, temperature.	Run compare_dosimetry_models.py and separate field, dose, and heating interpretations.	Manuscript Layer 1; README; compare_dosimetry_models.py.
11	Layer 2: electrodynamics	Resting V _m versus induced Delta V _m , RC/Schwan charging, organelle membranes, permeability and pore proxies.	Run compare_membrane_electrodynamics.py; vary pulse width and cell radius.	Manuscript Layer 2; electrodynamics.py; membrane-voltage figure.
12	Layers 3-4: signaling and repair	Ca ²⁺ fluxes and stores, ROS, ATP, osmotic stress, PS, calpain, annexin, actin, resealing.	Compare baseline, high-dose, Ca-limited, and repair-supported scenarios.	ion_transport.py; remodeling_repair.py; comparison examples and figures.
13	Layers 5-8: EV product	MVB/ILV pools, fusion, budding, apoptotic bodies, cargo, potency, injury mixture, purity, recovery, batch effects.	Contrast secretory, direct-loading, and injury-dominant regimes; defend which product is acceptable.	ev_release.py; cargo_potency.py; injury_quality.py; manufacturing_qc.py.
14	Integration and capstone	Cell-state modifiers, multiscale storyboard, solution spaces, calibration order, evidence, and honest uncertainty.	Present one submodel improvement, evidence table, validation plan, or teaching simulation.	storyboard_multilayer_overview.py; solution_space_analysis; model assumptions and evidence targets.

If a 15th week is available, split Week 9 into separate PBPK/PD and model-credibility weeks.

5. From prerequisites to the modular EV-biogenesis framework

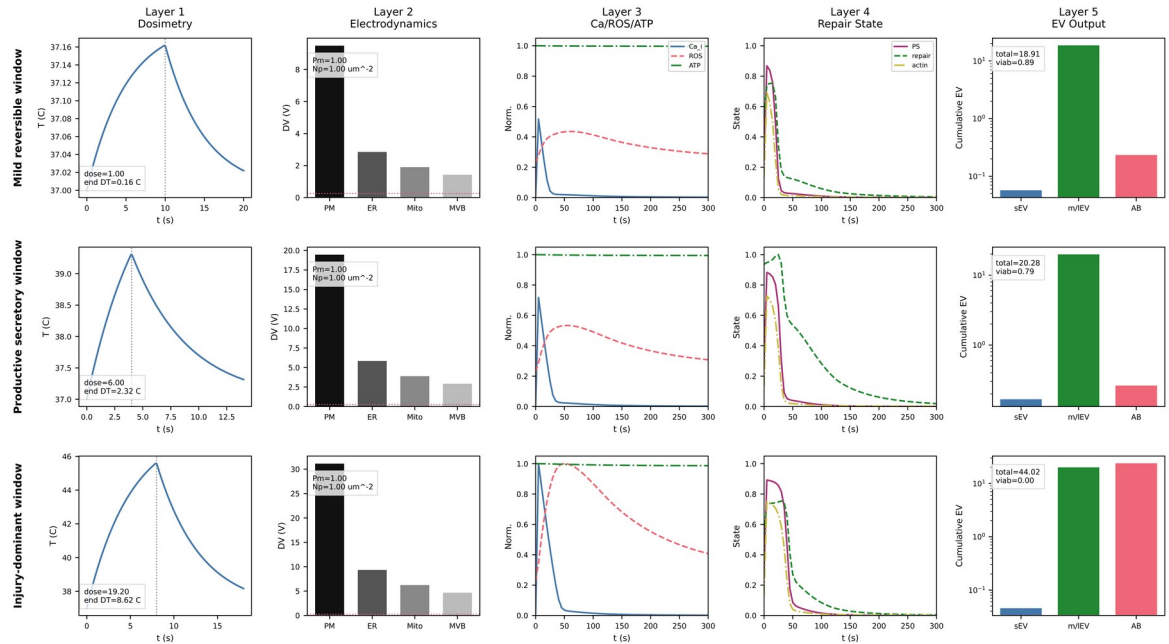
Plain-language organizing question: What happens between an electrical pulse delivered outside a cell and a useful, high-quality EV product collected hours later? Each layer answers one part of that question and passes a compact set of outputs to the next layer.

Module	What the module contributes	Background concept and classroom demonstration
1. Pulse delivery and dosimetry	Defines the external recipe and translates it into field, dose, energy, and temperature descriptors.	Prerequisite: units, external inputs, energy balances. Demo: compare dosimetry formulations.
2. Membrane/organelle electrodynamics	Converts the field into induced transmembrane voltages, conductance, permeability, and pore proxies.	Prerequisite: membrane potential, capacitance, Nernst/GHK. Demo: pulse-width and cell-size sensitivity.
3. Ion transport, Ca²⁺, ROS, ATP	Uses compartmental flux-balance ODEs to turn a brief electrical event into longer cellular trajectories.	Prerequisite: compartments, fluxes, pumps, channels, reactions. Demo: baseline vs high dose vs Ca-limited.
4. Remodeling and repair	Maps Ca ²⁺ and stress into PS exposure, calpain, annexin, actin remodeling, lysosomal repair, and resealing.	Prerequisite: Hill activation, feedback, competing processes. Demo: buffering, inhibition, repair support.
5. EV biogenesis and subtype release	Tracks MVB/ILV/docking pools, plasma-membrane budding, fusion, and apoptotic-body production.	Prerequisite: pool balances, exocytosis, rate laws. Demo: Rab/ESCRT-limited vs ceramide-favored vs injury-shifted.
6. Cargo, composition, and potency	Combines subtype-weighted protein, RNA, lipid, antigen, and direct-loading states into functional output.	Prerequisite: PK/PD-style effect mapping and saturation. Demo: endogenous sorting vs direct loading.
7. Injury, debris, and quality gate	Separates useful EVs from stressed cells, apoptosis, necrosis, debris, and aggregates using thresholds and mixtures.	Prerequisite: nonlinear switches, mixture accounting, model credibility. Demo: high particle count that fails quality.
8. Manufacturing, isolation, and QC	Converts cellular release into isolated yield, recovery, purity, batch consistency, scale, and an optimization objective.	Prerequisite: process mass balance, PBPK/flow intuition, constraints. Demo: yield-purity-viability tradeoff.
Cross-cutting cell-state modifiers	Scales parameters for cell type, disease state, membrane composition, Ca ²⁺ handling, metabolism, and stress susceptibility.	Prerequisite: variability, parameter distributions, sensitivity. Demo: resilient-secretory vs fragile-stress-biased cells.

Module names and interfaces follow `main.tex`, `README.md`, `docs/model_assumptions.md`, and the model files under `electro_exocytosis/models`.

The module contract students should use

- **Input:** what arrives from the experiment or the previous module?
- **State:** what quantities must be remembered over time?
- **Process:** what balance, rate law, threshold, or mapping changes those states?
- **Output:** what is passed downstream and what can be measured?
- **Evidence:** which experiment or source can constrain the parameters or test the behavior?
- **Limitation:** which assumption is most likely to fail for this cell type, timescale, or protocol?



Integrated storyboard across Layers 1-5. Use it after the individual modules, not before, so students can explain why mild, productive-secretory, and injury-dominant regimes diverge.

Source figure: /Users/aghoban/code/electro-exocytosis/figs/multilayer_storyboard.png

6. Labs, assessment, and capstone

Computational scaffolding

Use a no-code to guided-code to independent-analysis progression. In the first weeks, students predict curves and edit diagrams. By Weeks 3-5, they modify one equation or parameter in a prepared notebook. In the EV block, they run existing repository examples and explain the output before changing implementation details.

- **Lab 1 - One-box washout:** define volume, concentration, flow, time constant, and conserved amount.
- **Lab 2 - Two-compartment exchange:** change volume and permeability-surface area; explain equilibrium and timescale.
- **Lab 3 - Membrane potential:** compare Nernst single-ion predictions with a multi-ion GHK result.
- **Lab 4 - PK/PD:** connect a one-compartment concentration curve to an Emax or Hill effect curve.
- **Lab 5 - PBPK/PD:** add flow and tissue exchange; perform a one-parameter sensitivity analysis.
- **Lab 6 - electroexo Layers 1-2:** compare dosimetry and membrane charging across pulse protocols.
- **Lab 7 - Layers 3-4:** interpret Ca²⁺/ROS/ATP and repair trajectories across biological hypotheses.

- **Lab 8 - Layers 5-8:** classify a scenario as productive, direct-loading, or injury-dominant and justify the quality gate.

Example assessment mix

- 15% concept checks and figure explanations
- 35% guided computational labs
- 15% source/evidence critique
- 35% capstone proposal, simulation, presentation, and reflection

Strong capstone formats

- Improve one submodel and explain the biological assumption it replaces.
- Build one evidence-to-parameter table for a chosen layer and cell type.
- Design one validation experiment with measurable module inputs and outputs.
- Create one beginner-facing simulation or visualization that teaches a difficult concept correctly.
- Compare two cell-state parameterizations and identify which data would discriminate them.

Scientific honesty requirement: The current software is structurally complete but scientifically provisional. Students must distinguish a runnable model from a validated model and label placeholder parameters, assumptions, and evidence gaps explicitly.

7. Likely teaching pitfalls and how to avoid them

Pitfall	Teaching response
Starting with the full HH model	Use Nernst, GHK, capacitance, and one gating variable first. HH is optional structure, not the entry point.
Confusing resting V_m with induced ΔV_m	Name the source of each voltage every time: ionic gradients/permeability versus external-field charging.
Treating compartments as literal anatomy	Repeat that a compartment is a modeling decision tied to mixing and timescale assumptions.
Using PK/PD as a biological equivalence	Frame it as a reusable modeling pattern: input, disposition, state, and effect.
Stopping at total EV count	Require subtype, cargo/potency, viability, contamination, and process recovery in the final interpretation.
Hiding placeholder status	Pair every simulation with an assumptions/evidence box and a proposed validation measurement.

Pitfall	Teaching response
Showing the full framework too early	Preview the destination briefly, then revisit it only after the prerequisite vocabulary has been built.

These responses are consistent with the report's modular scope and the repository's explicit placeholder disclaimer.

8. Existing assets to reuse

Asset	Recommended role
outputs/course_authoring/ generate_electroexo_course_general_audience.mjs	A 10-session physiology-first slide sequence. Retain as the compressed seminar version.
outputs/course_authoring/ generate_electroexo_syllabus_ag_template.mjs	A 14-week syllabus scaffold. Update it with the page-level source assignments in this guide.
outputs/course_authoring/ generate_electroexo_course_slides.mjs	A more technical 10-session deck. Use after the general-audience concepts or as graduate enrichment.
examples/compare_*.py	Ready-made module demonstrations for dosimetry, electrodynamics, ion transport, repair, EV release, and downstream engineering.
examples/storyboard_multilayer_overview.py	Best integrated visual narrative after students have learned the individual modules.
examples/solution_space_analysis	Advanced exploration of pulse, waveform, conductivity, dosimetry, and cell-state parameter sweeps.
docs/model_assumptions.md and evidence targets	Required reading for limitations, calibration order, and capstone design.

Executable repository reviewed: /Users/aghorban/code/electroexo

9. Production roadmap for next year's course

Timing	Deliverable
T-9 to T-7 months	Confirm audience, contact hours, software policy, and whether the primary format is 14 weeks or the 10-session seminar.
T-7 to T-5 months	Create a canonical source folder, remove duplicates, and prepare a one-page reading guide for each week.
T-6 to T-4 months	Build five starter notebooks: compartments, membrane potential, PK/PD, PBPK/PD, and electroexo module runner.
T-4 to T-3 months	Revise the 14-week slide scaffold with the exact source pages and a common input-state-output-evidence visual language.
T-3 to T-2 months	Pilot Weeks 3, 5, and 12 with noncomputational learners; measure where vocabulary or plots become confusing.

Timing	Deliverable
T-2 to T-1 months	Finalize assessments, answer keys, accessibility checks, setup instructions, and a fallback no-install environment.
During delivery	Collect one-minute reflections on what each equation term means biologically and revise the next offering accordingly.

The roadmap is intentionally semester-neutral so it can be aligned to the actual 2027 offering date.

10. Canonical source roots and reading spine

Use these roots as the stable source-of-truth locations while assembling the course:

- Systems physiology: /Users/aghorban/Library/CloudStorage/OneDrive-Personal/Documents/04-Universities/MCW_MU/Courses/BIEN5700-SystemPhysiology
- Computational systems biology:
/Users/aghorban/Library/CloudStorage/OneDrive-Personal/Documents/04-Universities/MCW_MU/Courses/BIEN6931-TopicsInBiomedicalEngineering/end of the semester back up/BIEN 6931 701 Topics in Biomedical Engr - 5132018 - 1103 PM
- EV-biogenesis manuscript and figures: /Users/aghorban/code/electro-exocytosis
- Executable electroexo model and examples: /Users/aghorban/code/electroexo

Minimal student reading spine

- 01homeostasis.pdf - homeostasis, membrane transport, exocytosis, and membrane electrical potentials.
- Dash_Lect1_Intro_Sim_Biol_Syst_Spring_2018.pdf - what computational systems biology is and how models are developed.
- Introduction to Compartmental Modeling (1).pdf - the core mass-balance examples.
- Numerical_methods_Spring_2018.pdf - solver intuition and numerical error.
- Compartmental_Kinetic_PKPD_Modeling.pdf and PBPK_spring_2018.pdf - staged exposure/effect and physiology-based compartments.
- Electrochemical_gradient_4_2_2018.pdf and Resting_membrane_potential_4_2_2-18.pdf - charged transport and voltage.
- Dash_Modeling Biochemical Reaction Systems.pdf - rate laws, energy coupling, and enzyme kinetics.
- The manuscript module map plus README.md and docs/model_assumptions.md - the final EV framework and its limitations.

Immediate next build

Recommended next artifact: Update the existing 14-week syllabus and slide authoring pipeline with these exact page assignments, then build the first four starter notebooks. That sequence locks the course architecture before substantial slide production begins.